2.2 수열의 극한

- sequence is a function $f: \mathbb{N} \to \mathbb{R}$

- $\underbrace{(a_n) \to a}_{\text{convergence}}$: $a_n \in \underbrace{(a-\varepsilon, a+\varepsilon)}_{\varepsilon\text{-neighbourhood}}$ $N_\varepsilon(a)$

$\left(\forall \varepsilon > 0,\ \exists n_0 \in \mathbb{N}\ \underbrace{\forall n \geq n_0}_{\text{"eventually"}} \right)$

in other words $|a_n - a| < \varepsilon$

- 수열의 극한값은 (존재한다면) 유일하다.

2.3 극한에 대한 정리

- $\underbrace{\text{converging}}_{a_n \in (a-\varepsilon, a+\varepsilon)\ \forall n \geq N}$ sequence is $\underbrace{\text{bounded}}_{a_n \in [-M, M]\ \forall n \in \mathbb{N}}$

$\because$

$M := \max\{\underbrace{|a_1|, \dots, |a_N|}_{\varepsilon\text{-근방 외부}}, \underbrace{|a|+\varepsilon}_{\varepsilon\text{-근방 내부}}\}$

- 수렴하는 수열에 대해서는 사칙연산 + 부등식 연산이 수행 가능하다.

suppose $(a_n) \to a$, $(b_n) \to b$

사칙연산

① $(ca_n) \to ca$

② $(a_n + b_n) \to a+b$

③ $(a_n b_n) \to ab$

④ $(a_n / b_n) \to a/b$ $(b \neq 0)$

부등식

① $0 \leq a_n$ $(\forall n \in \mathbb{N})$ then $0 \leq a$

② $a_n \leq b_n$ (") then $a \leq b$

③ $a_n \leq c \leq b_n$ (") then $a \leq c \leq b$

$\because$ 사칙연산 ④

$\boxed{\left(\frac{1}{b_n}\right) \to \left(\frac{1}{b}\right) \quad (b > 0)}$

$\left| \frac{1}{b_n} - \frac{1}{b} \right| = \frac{|b_n - b|}{|b_n||b|}$

$< \frac{2}{b} \cdot \frac{1}{b} \cdot \varepsilon$

$\forall n \geq N$

$b_n \in \underbrace{\left(\frac{1}{2}b, \frac{3}{2}b\right)}_{b_n\text{은 양수이다}}$

$\underbrace{|b_n - b|}_{b\text{와 가깝다.}} < \varepsilon \quad (\forall \varepsilon > 0)$

$\because$ 부등식 ①

suppose $a < 0$ then $\forall n \geq N$

$a_n \in (a-\varepsilon, \underbrace{a+\varepsilon}_{<0})$

2.4 단조수렴정리와 무한급수

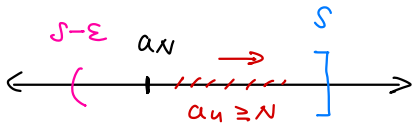
- monotone sequence
  - increasing $a_n \le a_{n+1}$ $(\forall n \in \mathbb{N})$
  - decreasing $a_n \ge a_{n+1}$ $(\forall n \in \mathbb{N})$

- 단조수렴정리

$\underbrace{\text{bounded}}_{a_n \in [-M, M]}$ & $\underbrace{\text{monotone}}_{a_n \le a_{n+1}}$

sequence $\underbrace{\text{converges}}_{a_n \in (a-\varepsilon, a+\varepsilon)}$ (to, supremum / infimum)

$\because$



$\underbrace{s-\varepsilon}_{\varepsilon\text{-근방}} \underbrace{<}_{\text{상계 아님}} a_N \underbrace{\le}_{\text{단조 증가}} a_n \underbrace{< s}_{\text{상한}}$

- $\underbrace{\sum_{n=1}^{\infty} b_n}_{\text{infinite series}} = B$ equals to $(s_m) \longrightarrow B$

partial sum : $s_m = (b_1 + b_2 + \cdots + b_m)$

- for $\underbrace{\text{non-negative series}}_{a_n \ge 0}$, $\underbrace{\text{bounded (above)}}_{\sum_{n=1}^{\infty} a_n < \infty} \Longleftrightarrow \underbrace{\text{convergence}}_{\sum_{n=1}^{\infty} a_n = B}$

$\sum_{n=1}^{\infty} \frac{1}{n^2} < \infty$

$\because$

$$s_m = 1 + \frac{1}{2\cdot 2} + \frac{1}{3\cdot 3} + \cdots + \frac{1}{m^2}$$

$$< 1 + \frac{1}{2\cdot 1} + \frac{1}{3\cdot 2} + \cdots + \frac{1}{m(m-1)}$$

$$= 1 + \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{m-1} - \frac{1}{m}\right)$$

$$< 2$$

$\sum_{n=1}^{\infty} \frac{1}{n} = \infty$

$\because$

$$s_{2^k} = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \cdots + \left(\frac{1}{2^{k-1}+1} + \cdots + \frac{1}{2^k}\right)$$

$$> 1 + \frac{1}{2} + \underbrace{\left(\frac{1}{4} + \frac{1}{4}\right)}_{1/2} + \cdots + \underbrace{\left(\frac{1}{2^k} + \cdots + \frac{1}{2^k}\right)}_{1/2}$$

$$= 1 + k \cdot \frac{1}{2}.$$

- 코시 응집판정법 $\underbrace{b_n \ge b_{n+1}}_{\text{감소수열}} \ge \cdots \ge \underbrace{0}_{\text{non-negative}}$ $(\forall n \in \mathbb{N})$

then $\sum_{n=1}^{\infty} b_n < \infty$ iff $\underbrace{\sum_{n=1}^{\infty} 2^n \cdot b_{2^n}}_{b_1 + 2b_2 + 4b_4 + \cdots} < \infty$

(pf) $b_4 + b_4 + b_4 + b_4$
$\ge b_4 + b_5 + b_6 + b_7$
$\ge b_8 + b_8 + b_8 + b_8$

(cf) $\lim \left(\frac{a_{2^{n+1}}}{a_{2^n}}\right)$
- $< \frac{1}{2}$: 수렴
- $> \frac{1}{2}$: 발산
- $= \frac{1}{2}$: 판정불가

$\sum_{n=1}^{\infty} \frac{1}{n^p} < \infty$ iff $p > 1$

$\because$ $\sum_{n=1}^{\infty} 2^n \cdot \frac{1}{2^{np}} < \infty$ $\underbrace{\text{iff}}_{\text{등비급수}}$ (ratio) $= \left(\frac{2}{2^p}\right) < 1$

2.5 부분수열과 볼차노-바이어슈트라스 정리

- 수열 (a_n)의 subsequence (a_{n_k})

  자연수 수열 $n_1 < n_2 < n_3 \cdots$.

- if $(a_n) \to a$ then $\forall (a_{n_k}) \to a$

  부분수열 극한값 동일

$\because$ if $k \geq N$ where $|a_k - a| < \varepsilon$ then $n_k \geq N$

극한값과 가까움.

- $(b^n) \to 0$ iff $|b| < 1$
- $0 < b < 1$ then $\begin{cases} (b^n) \to 0 \\ \left(\frac{1}{b^n}\right) \to \infty \end{cases}$

$\because$ $(b^n) \longrightarrow l$ (단조수렴정리)

$(b^{2n}) \longrightarrow \boxed{l}$ (부분수열)

$(b^n \cdot b^n) \longrightarrow \boxed{l^2}$ (사칙연산)

$l = 0$

- $(a_n) = (0, 1, 0, 1, 0, 1 \cdots)$ 는 발산한다.

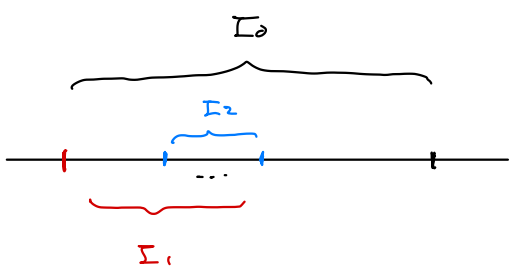
$\because$ $(a_{2n-1}) \to 0$

$(a_{2n}) \to 1$

Bolzano - Weierstrass Thm.

- 유계수열은 수렴하는 부분수열을 가진다.

$a_n \in [-M, M]$



$\because$

- $I_0 = [-M, M]$
- I_k를 2등분하여 (a_n)의 항을 무한히 많이 가지는 I_{k+1}을 선택한다.
- $I_0 \supseteq I_1 \supseteq I_2 \supseteq I_3 \cdots \supseteq \bigcap_{k=1}^{\infty} I_k$

  $\quad\ \ni a_{n_1} \quad \ni a_{n_2} \quad \ni a_{n_3} \quad \ni a_{n_k} \qquad\qquad \ni x$ (축소구간정리)

- $|a_n - x| \leq \underbrace{\text{length}(I_k)}_{\to 0}$

2.6 코시 수열 판정법

Sequence converges iff $\underbrace{\text{Cauchy Sequence}}$

$|a_n - a_m| < \varepsilon$

$(\forall n, m \geq N)$

Suppose (x_n) is Cauchy Sequence

① (x_n) is 유계 $\qquad |x_{n \geq N}| \leq |x_N| + 1$

② (x_n) has 수렴하는 부분수열 $(x_{n_k}) \longrightarrow x$

③ $(x_n) \longrightarrow x$

$\because$ $|x_n - x| \leq \underbrace{|x_n - x_{n_k}|}_{\text{코시 수열}} + \underbrace{|x_{n_k} - x|}_{\text{수렴하는 부분수열}} < \varepsilon$

2.7 무한급수의 성질

사칙연산 suppose $\sum_{k=1}^{\infty} a_k = A$, $\sum_{k=1}^{\infty} b_k = B$

then ① $\sum_{k=1}^{\infty} (ca_k) = cA$

② $\sum_{k=1}^{\infty} (a_k + b_k) = A + B$

코시 수렴판정법 $\sum_{k=1}^{\infty} a_k$ converges iff $\underbrace{|a_{m+1} + \cdots + a_n|}_{\forall n > m \geq N} < \underbrace{\varepsilon}_{\varepsilon > 0}$

$S_n - S_m$

Note. $\sum_{k=1}^{\infty} a_k$ converges then $(a_k) \to 0$

Note. (converse) Harmonic Series $\sum_{n=1}^{\infty} \frac{1}{n} = \infty$ but $\left(\frac{1}{n}\right) \to 0$

비교판정법 $\forall k \in \mathbb{N}$ $0 \leq a_k \leq b_k$

$\because$ $0 \leq |a_{m+1} + \cdots + a_n|$

$\leq |b_{m+1} + \cdots + b_n|$

then ① $\sum_{k=1}^{\infty} b_k$ converges then $\sum_{k=1}^{\infty} a_k$ converges

② $\sum_{k=1}^{\infty} a_k$ diverges then $\sum_{k=1}^{\infty} b_k$ diverges

---

등비급수

$$\sum_{k=0}^{\infty} ar^k = \underbrace{a + ar + \cdots + ar^{m-1}}_{a \cdot \left(\frac{1-r^m}{1-r}\right)} + ar^m + \cdots = \underbrace{\frac{a}{1-r}}_{\text{iff } |r| < 1}$$

절대수렴과 조건수렴 (positives) $p_k = \max(a_k, 0)$

(negatives) $q_k = \min(a_k, 0)$

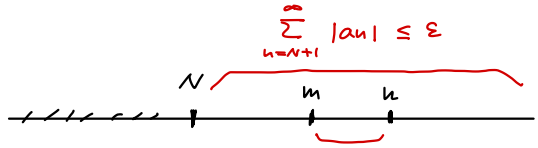
$\sum_{k=1}^{\infty} a_k$ converges then

- 절대수렴 $\Longrightarrow$ $\sum_{k=1}^{\infty} |a_k| < \infty$ i.e $\sum_{k=1}^{\infty} p_k < \infty$, $\sum_{k=1}^{\infty} |q_k| < \infty$
- 조건수렴 $\Longrightarrow$ $\sum_{k=1}^{\infty} |a_k| = \infty$ i.e $\sum_{k=1}^{\infty} p_k = \infty$, $\sum_{k=1}^{\infty} q_k = -\infty$

절대수렴 $\sum_{k=1}^{\infty} |a_k|$ converges then $\sum_{k=1}^{\infty} a_k$ converges

(pf1) $\sum_{k=1}^{\infty} p_k$, $\sum_{k=1}^{\infty} |q_i| \le \sum_{k=1}^{\infty} |a_k|$ therefore $\sum_{k=1}^{\infty} \underbrace{(p_k + q_k)}_{= a_k}$ converge

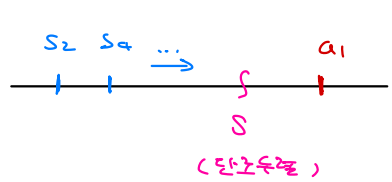
(pf2)



$\varepsilon > |a_{m+1}| + \cdots + |a_n|$ (절대수렴)

$\ge |a_{m+1} + \cdots + a_n|$

교대급수 판정법



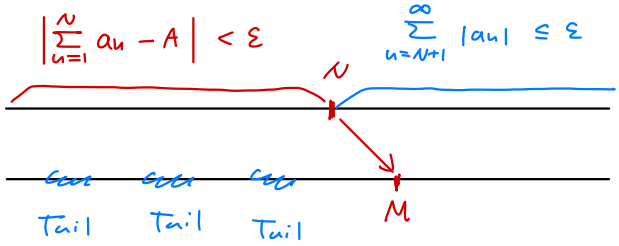
$$\begin{cases} a_1 \ge \cdots \ge a_n \ge a_{n+1} \ge \cdots \\ (a_n) \to 0 \end{cases}$$

then $a_n \ge 0$ and $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ converges

---

재배열 $(\forall k \in \mathbb{N})$ $b_k = a_{\underbrace{f(k)}_{\text{bijection } \mathbb{N} \to \mathbb{N}}}$ then (b_k) is rearrangement of (a_k)

절대수렴하는 급수의 재배열 $\Longrightarrow$ 극한값 동일



$$\left| (b_1 + \cdots + b_M) - (a_1 + \cdots + a_N) \right| + \left| (a_1 + \cdots + a_N) - A \right|$$

(a_n)의 Tail $\qquad$ (a_n)의 Main Body

조건수렴하는 급수의 재배열 $\Longrightarrow$ $\forall r \in \mathbb{R}$

algorithm
$$\begin{cases} \text{while } (\text{부분합} \le r) \text{ then add } p_i \\ \text{while } (\text{부분합} \ge r) \text{ then add } q_i \end{cases}$$

$a_n \to 0$ therefore $(p_n) \to 0$ and $(q_i) \to 0$ (오차항)

---

alternating harmonic series

$$\begin{array}{rl} S = & 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \frac{1}{8} \\ + \quad \frac{1}{2} S = & \qquad \frac{1}{2} \qquad - \frac{1}{4} \qquad + \frac{1}{6} \qquad - \frac{1}{8} \\ \hline \frac{3}{2} S = & 1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} \end{array}$$

2.8 이중급수와 부분수열의 합.

" in short (조건수렴)

$$\sum_{i=1}^{\infty}\sum_{j=1}^{\infty} a_{ij} \neq \sum_{j=1}^{\infty}\sum_{i=1}^{\infty} a_{ij} \text{ "}$$

$$\begin{array}{c|c} & \begin{array}{cccc} -1 & -\frac{1}{2} & -\frac{1}{4} & \cdots \end{array} \\ \hline \begin{array}{c} \odot \\ \odot \\ \odot \\ \ddots \end{array} & \begin{bmatrix} -1 & \frac{1}{2} & \frac{1}{4} & \\ & -1 & \frac{1}{2} & \vdots \\ & & -1 & \\ & \cdots & & \end{bmatrix} \end{array}$$

---

if $\sum_{i=1}^{\infty}\sum_{j=1}^{\infty} |a_{ij}|$ converges

$d_k = a_{1,k-1} + \cdots + a_{k-1,1}$

then (by 절대수렴) $\lim_{n\to\infty} S_{nn} = \sum_{i=1}^{\infty}\sum_{j=1}^{\infty} a_{ij}$

$$= \sum_{j=1}^{\infty}\sum_{i=1}^{\infty} a_{ij} = \sum_{k=2}^{\infty} d_k$$

---

let $S_{mn} = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}$, $t_{mn} = \sum_{i=1}^{m}\sum_{j=1}^{n} |a_{ij}|$

then ① (t_{nn}) 단조수열.

절댓값 L 들라이스가 작아지는 N 존재.

② (S_{nn}) 코시수열

S_{mm}이 S에 가까워지는 N 존재

③ $|S_{NN} - S| + |S_{mn} - S_{NN}|$ therefore $(S_{mn}) \longrightarrow S$

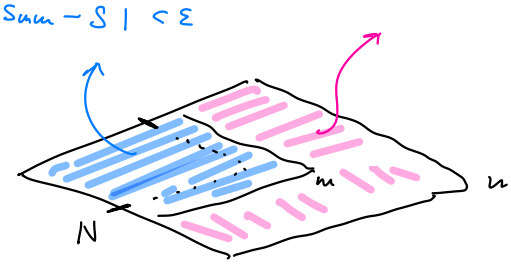
$< \varepsilon$ $\quad \leq t_{nn} - t_{NN}$

or $\leq t_{mn} - t_{NN}$

④ $\sum_{i=1}^{\infty}\sum_{j=1}^{\infty} a_{ij} = \lim_{m\to\infty}\lim_{n\to\infty} S_{mn}$

$t_{nn} - t_{mm} < \varepsilon$

$|S_{mm} - S| < \varepsilon$



코시 곱

$$\left(\sum_{i=1}^{\infty} a_i\right)\left(\sum_{j=1}^{\infty} b_j\right) = \underbrace{a_1b_1}_{d_2} + \underbrace{(a_1b_2 + a_2b_1)}_{d_3} + \underbrace{(a_1b_3 + a_2b_2 + a_3b_1)}_{d_4} + \cdots$$

Suppose $\sum_{i=1}^{\infty} a_i = A$ (absolute), $\sum_{j=1}^{\infty} b_j = B$ (absolute)

then $\sum_{i=1}^{\infty}\sum_{j=1}^{\infty} |a_ib_j|$ converges

and (...) therefore $\sum_{i=1}^{\infty}\sum_{j=1}^{\infty} a_ib_j = \sum_{j=1}^{\infty}\sum_{i=1}^{\infty} a_ib_j = \sum_{k=2}^{\infty} d_k = AB$

2.9 리뷰

power series

$$(a_0 + a_1x + a_2x^2 + \cdots)(b_0 + b_1x + b_2x^2 + \cdots)$$

$$= \underbrace{a_0b_0}_{d_0} + \underbrace{(a_0b_1 + a_1b_0)}_{d_1}x + \underbrace{(a_0b_2 + a_1b_1 + a_2b_0)}_{d_2}x^2 + \cdots$$

코시곱의 수렴.

	$\sum a_n = A$, $\sum b_n = B$	코시곱 $\sum d_k$
Cauchy	둘다 절대수렴	절대수렴.
Meutens	하나는 절대수렴, 하나는 그냥 수렴	수렴
Abel	둘 다 수렴	수렴하면 그 값은 AB